Chapter 7

Linear Hyperbolic Systems

7.1 The Cauchy Problem

For $\mathbf{u} : \mathbb{R} \times [0, \infty) \to \mathbb{R}^m$, consider the system-Cauchy problem

$$\begin{aligned} \mathbf{u}_t + A\mathbf{u}_x &= 0, \qquad x \in \mathbb{R},\ t > 0 && (7.1) \\ \mathbf{u}(x, 0) &= \mathbf{u}_0(x). && (7.2) \end{aligned}$$

where $A \in \mathbb{R}^{m\times m}$ is a square matrix. The initial conditions are given by the vector-valued function $\mathbf{u}_0 : \mathbb{R} \to \mathbb{R}^m$.

By definition of a hyperbolic system (cf. Section 5.2), we can diagonalize the matrix A. This means there is an invertible matrix $R \in \mathbb{R}^{m\times m}$, such that $R^{-1}AR = \Lambda$, where $\Lambda = diag(\lambda_1, \ldots, \lambda_m)$, and the λ_k are (real) eigenvalues of the matrix A. We write the system (7.1)

$$\mathbf{u}_t + R\Lambda R^{-1}\mathbf{u}_x = 0.$$

Setting $R^{-1}\mathbf{u} =: \mathbf{w} = (w^{(1)}, \ldots, w^{(m)})^T$, we obtain

$$\begin{aligned} \mathbf{w}_t + \Lambda\mathbf{w}_x &= 0, \qquad x \in \mathbb{R},\ t > 0, && (7.3) \\ \mathbf{w}(x, 0) &= \mathbf{w}_0(x), && (7.4) \end{aligned}$$

where $\mathbf{w}_0 = R^{-1}\mathbf{u}_0$. The equations are thus decoupled. (Note that Λ is a diagonal matrix!) Thus (7.3), (7.4) means that for $k = 1, \ldots, m$,

$$\begin{aligned} w_t^{(k)} + \lambda_k w_x^{(k)} = 0, \qquad . \\ w^{(k)}(x, 0) = w_0^{(k)}(x). \end{aligned} \tag{7.5}$$

Each of these equations is a linear advection equation. The solution to the kth equation is simply $w_0^{(k)}(x - \lambda_k t)$. The solution to the original system is thus a superposition of these problems. Denote the eigenvectors of A as $\mathbf{r}_k$ for $k = 1, \ldots m$. Then we can write the matrix $R = (\mathbf{r}_1|\ldots|\mathbf{r}_m)$, where the $|$ symbol means that the column vectors are concatenated to form the columns of the matrix R, and one has:

$$\mathbf{u}(x, t) = R\mathbf{w}(x, t) = \sum_{k=1}^{m} \mathbf{r}_k w_k = \sum_{k=1}^{m} \mathbf{r}_k w_k(x - \lambda_k t) \tag{7.6}$$

7.2 Upwind Scheme for the System

A numerical scheme for the system is obtained if we use an upwind discretization for each k in (7.5). Note that we have to use the correct upwind direction, depending on the sign of the

eigenvalues. Define $\lambda_p^+ = \max(0, \lambda_p)$ and $\lambda_p^- = \min(\lambda_p, 0)$. Then set $\Lambda^\pm = diag(\lambda_1^\pm, \ldots, \lambda_m^\pm)$. Note that $\Lambda = \Lambda^+ + \Lambda^-$. A scheme for the system is obtained by setting

$$\frac{\mathbf{w}_i^{n+1} - \mathbf{w}_i^n}{\tau} + \Lambda^+ \frac{\mathbf{w}_i^n - \mathbf{w}_{i-1}^n}{h} + \Lambda^- \frac{\mathbf{w}_{i+1}^n - \mathbf{w}_i^n}{h} = 0.$$

We may write this

$$\frac{\mathbf{w}_i^{n+1} - \mathbf{w}_i^n}{\tau} + \Lambda \frac{\mathbf{w}_{i+1}^n - \mathbf{w}_{i-1}^n}{2h} - |\Lambda| \frac{\mathbf{w}_{i+1}^n - 2\mathbf{w}_i^n + \mathbf{w}_{i-1}^n}{2h} = 0.$$

where $|\Lambda| := diag(|\lambda_1|, \ldots, |\lambda_m|)$. Transforming back to the original variable we obtain

$$\frac{\mathbf{u}_i^{n+1} - \mathbf{u}_i^n}{\tau} + A \frac{\mathbf{u}_{i+1}^n - \mathbf{u}_{i-1}^n}{2h} - |A| \frac{\mathbf{u}_{i+1}^n - 2\mathbf{u}_i^n + \mathbf{u}_{i-1}^n}{2h} = 0 \tag{7.7}$$

where $|A| = R|\Lambda|R^{-1}$.

7.3 Stability for Systems

The equivalent of the generic difference scheme (4.7) for systems of equations can be written

$$\sum_{m=-r}^{s} \widetilde{D}_m \mathbf{u}_{i+m}^{n+1} = \sum_{m=-p}^{q} D_m \mathbf{u}_{i+m}^n, \tag{7.8}$$

where now $\widetilde{D}_m, D_m \in \mathbb{R}^{m \times m}$.

Example 7.1 The scheme (7.7) can be written in the form (7.8) with

$$\begin{aligned} D_{-1} &= \frac{\tau}{2h}(A + |A|) \\ D_0 &= I - \frac{\tau}{h}|A| \\ D_1 &= \frac{\tau}{2h}(-A + |A|) \\ \widetilde{D}_0 &= I \end{aligned}$$

All other coefficients are zero.

The analysis of section 4.4 is extended in a rather straightforward manner. Each component of the solution can be represented using the complex-exponential basis that was introduced in Section (4.4):

$$\sum_{m=-r}^{s} \widetilde{D}_m \sum_{k=0}^{N-1} \mathbf{c}_k^{n+1} \phi_{k,i+m} = \sum_{m=-p}^{q} D_m \sum_{k=0}^{N-1} \mathbf{c}_k^n \phi_{k,i+m} \qquad i = 0, \ldots, N-1$$

where $\mathbf{c}_k = (c_{k,1}, \ldots, c_{k,m})^T$. Rearranging the sums, we have

$$\sum_{k=0}^{N-1} \sum_{m=-r}^{s} \widetilde{D}_m \mathbf{c}_k^{n+1} \mu_k^m \phi_{k,i} = \sum_{k=0}^{N-1} \sum_{m=-p}^{q} D_m \mathbf{c}_k^n \mu_k^m \phi_{k,i} \qquad i = 0, \ldots, N-1.$$

Here μ_k are the eigenvalues defined in Section 4.4 (cf. eq. (4.21)). Each component corresponds to an expansion in the same basis function on both sides. Since expansion in a basis is unique

(i.e. the coefficients are unique) we have

$$\begin{aligned} \sum_{m=-r}^{s} \widetilde{D}_m \mathbf{c}_k^{n+1} \mu_k^m &= \sum_{m=-p}^{q} D_m \mathbf{c}_k^n \mu_k^m \\ \iff \underbrace{\left(\sum_{m=-r}^{s} \widetilde{D}_m \mu_k^m \right)}_{\widetilde{M}(h,\tau,k)} \mathbf{c}_k^{n+1} &= \underbrace{\left(\sum_{m=-p}^{q} D_m \mu_k^m \right)}_{M(h,\tau,k)} \mathbf{c}_k^n \end{aligned}$$

Taking the norm on both sides, we obtain

$$\left\| \mathbf{c}_k^{n+1} \right\| = \left\| \widetilde{M}^{-1} M \mathbf{c}_k^n \right\| \leq \left\| \widetilde{M}^{-1} M \right\| \left\| \mathbf{c}_k^n \right\| \Rightarrow \frac{\left\| \mathbf{c}_k^{n+1} \right\|}{\left\| \mathbf{c}_k^n \right\|} \leq \|\mathcal{R}\|$$

where $\mathcal{R} := \widetilde{M}^{-1} M$. By recursion we have

$$\frac{\left\| \mathbf{c}_k^n \right\|}{\left\| \mathbf{c}_k^0 \right\|} \leq \|\mathcal{R}^n\| \leq \|\mathcal{R}\|^n$$

Hence, $\|\mathcal{R}\| \leq 1$ is certainly sufficient for stability.